John P. Ortiz

Staff Scientist
National Security Earth Science Group (EES-17)
Earth & Environmental Sciences Division
Los Alamos National Laboratory

johnportiz14.github.io
jportiz@lanl.gov
+1 (505) 665-5901

Areas of Specialization

multiphase flow & transport • hydrogeology • fracture-matrix interactions • computational physics modeling • reactive transport • planetary science • geomechanics

Education

2024 Johns Hopkins University (JHU)
Ph.D. in Environmental Health & Engineering
Dissertation: Subsurface Flow and Transport Processes with Applications to Methane Variations on Mars
Advisor: Dr. Harihar Rajaram

2017 New Mexico Institute of Mining and Technology (NMT)
M.Sc. in Hydrology
Thesis: The role of fault-zone architectural elements and basal altered zones on downward pore pressure propagation and induced seismicity in the crystalline basement
Advisor: Dr. Mark Person

2015 Dartmouth College
B.A. in Earth Sciences with Honors
Honors Thesis: Quantifying regional sediment flux from observations of nearshore morphology in the Columbia River Littoral Cell
Advisor: Dr. W. Brian Dade

Planetary Exploration Mission Experience

2023 NASA TLS-SAM Experiment Proposal, Co-author
Co-authored proposal with members of TLS-SAM (Tunable Laser Spectrometer within the Sample Analysis at Mars) team proposing strategic timing of atmospheric sample experiments for Mars Science Laboratory (MSL) *Curiosity* rover to perform in order to constrain sub-diurnal methane variations at Gale crater, Mars. MSL *Curiosity* successfully executed one of the proposed experiments 23 September, 2023, and a second on 10 December, 2023. Authors: Daniel Lo (Univ. of Michigan), Sushil Atreya (Univ. of Michigan), Scot Rafkin (Southwest Research Institute), Jorge Pla-García (Centro de Astrobiología y Instituto Nacional de Técnica Aeroespacial; CSIC-INTA), John Moores (York University), Daniel Viúdez-Moreiras (CSIC-INTA), **John Ortiz**.

Honors & Awards

2026 Lindau Nobel Laureate Meetings Fellow
Selected as a 2026 University of California National Laboratories 75th Lindau Nobel Laureate Meetings Fellow (https://www.lindau-nobel.org/), which covers expenses to attend the interdisciplinary meeting to be held from June 28 to July 3 in Lindau, Germany. Was nominated by EES Division and selected as 1 of 4 fellows from LANL (see article in News Coverage/Web Articles).
As part of the Lindau meeting, I was also selected to present a talk to Nobel Laureates and young scientists during the *Next Gen Science* session (see Conference & Meetings Talks and Lindau Meeting Programme).

2026 Editor's Choice, ACS Publications (*ACS Omega*)
Article selected for ACS Editors' Choice®, a collection designed to feature new scientific articles of broad public interest. The article "Linking Laboratory and Field Observations of Carbon Dioxide Transport at the Nevada National Security Sites" (see Publications & Presentations) was selected based on recommendations from the Editors.

2026 Top Viewed Article, Wiley Publishing (*Journal of Geophysical Research: Planets*)
Amongst articles published in *Journal of Geophysical Research: Planets* between January 1, 2024 and December 31, 2024, the article "Sub-Diurnal Methane Variations on Mars Driven by Barometric Pumping and Planetary Boundary Layer Evolution" (see Publications & Presentations) was in the top 10% of most-viewed papers published in the journal in the 12-months post online publication.

2026 "Spot" Performance Award (July), Los Alamos National Laboratory
From the award: "For organizing the 'EES Day' event."

2025 Director's Postdoctoral Fellowship Awardee, Los Alamos National Laboratory
Awarded by Winter 2025 Postdoc Fellow Quarterly Review. One of the two types of Postdoctoral Fellow appointments, supporting the most promising among the Laboratory postdoc population. Selections are made based on academic and research accomplishments, the strength of the proposed research, as well as their potential impact at the Laboratory. Typically, up to seven appointments are awarded quarterly.

2020 R&D 100 Award Winner (*Amanzi-ATS*), Contributor
Contributed to the Amanzi-ATS multiphase flow and transport simulator project by augmenting code verification and benchmark tests in addition to maintaining user guide documentation. The code won an R&D 100 Award in September 2020. rdworldonline.com

2024 Oscar Academy Awards, Background Actor
Background actor (uncredited) in 7-time Oscar Award winning film, "Oppenheimer".

2018 Top 20 Most Downloaded Recent Papers, Wiley Publishing (*Geofluids*)
Amongst articles published between July 2016 and June 2018, the article "Exploring the potential linkages..." (see Publications & Presentations) was in the top 20 for number of downloads in the 12-months post online publication.

2008 Microsoft Office Worldwide Competition
Ranked #7 in the world in Microsoft Excel at the 2008 Microsoft Office Worldwide Competition in Hawaii of 56,000 initial entrants (www.betheworldchamp.com). Qualified as the Ireland National Champion (of 436 entrants).

Service Activities and Outreach

Invited Talks

2026 "Recap of the 75th Lindau Nobel Laureate Meeting", EES Division Leadership Team Meeting. Los Alamos, NM. 9 September, 2026.

2026 "From Manhattan to Mars: Applying Models of Subsurface Radionuclide Gas Seepage From Nuclear Testing to Understand a Martian Methane Mystery", 75th Lindau Nobel Laureate Meeting; *Next Gen Science Presentations*. Lindau, Germany. 1 July, 2026.
(see Lindau Meeting Programme)

2025 "Sub-diurnal methane variations on Mars driven by barometric pumping and planetary boundary layer evolution", NASA Sample Analysis at Mars (SAM) Team Science Meeting. 4 April, 2025

2023 "The Mars Underground: Characterizing subsurface methane seepage on the Red Planet", Los Alamos Geological Society monthly meeting. 18 April, 2023.

2023 "The Mars Underground: Characterizing subsurface methane seepage on the Red Planet", New Mexico Bureau of Geology Seminar Series, New Mexico Bureau of Geology and Mineralogical Resources. 7 April, 2023.

Conference Session Chair

2026 *Session Co-Convener and Session Chair, AGU Fall Meeting 2026*
"(P031) The New Mars Underground IX: Maximizing Martian Subsurface Science Return", San Francisco, CA, USA.

2025 *Primary Session Convener and Oral Session Co-Chair, AGU Fall Meeting 2025*
"The New Mars Underground VIII: Maximizing Mars Science Return", New Orleans, LA, USA.

2024 *Session Co-Convener and Oral Session Co-Chair, AGU Fall Meeting 2024*
"The New Mars Underground: Fluids, Volatiles, and the Future of Mars Exploration", Washington DC, USA.

Panels

2026 "Alumni Insights Panel: Working in a National Laboratory", Johns Hopkins University Doctoral Life Design Studio. Online. 10 September, 2026.

Public Outreach

August 2025 *LANL Student Symposium Judge*
Served as a judge of student research project posters, providing feedback to undergraduate

and graduate student presenters.

April 2024 *Los Alamos Middle School Guest Lecturer*
Delivered a series of five presentations over two days on measuring atmospheric methane on Mars with the *Curiosity* rover to several 7th and 8th grade classes of Life Sciences and Astronomy students at Los Alamos Middle School.

April 2023 *Los Alamos Geological Society monthly meeting*
Delivered an invited talk ("The Mars Underground: Characterizing subsurface methane seepage on the Red Planet"; see Invited Talks) about predicting methane abundance variations at Gale crater, Mars to members of the Los Alamos Geological Society for their monthly meeting.

Advisory Roles

2018 - 2019 *LANL Student Programs Advisory Committee (SPAC)*
Served a 1-year appointment as Graduate Student Representative to a committee that advises Los Alamos National Laboratory's Student Programs Office on matters related to the hiring and general well-being of student employees.
https://int.lanl.gov/employees/education/spac.shtml

Journal Reviewer

Journal of Environmental Radioactivity
Geophysical Research Letters
Icarus
Journal of Geophysical Research: Solid Earth
Hydrogeology Journal

Research Funding

To date, I have independently secured over $560,000 in research funding, supporting my work on both individual and collaborative projects, first as student and then as an early-career researcher.
The asterisk (*) indicates a grant for which I was not an official PI due to either graduate student or postdoc status (per rules of the grant), but on which I was the lead writer and researcher.

2025 *Director's Postdoctoral Fellow Research Grant
Re-caging the Dragon: Geomechanical insights to mitigate UNE containment failure risks.
This project improves modeling of underground nuclear explosion (UNE) test containment by investigating how extreme pressures and rock behavior influence stress cage formation. Findings will enhance nuclear test readiness and national security. Fully funds postdoc for one year (100% FTE) and provides 66% FTE for postdoc in second year. Project: 20251154PRD2. PI: Esteban Rougier; **Co-I/Fellow: John Ortiz**. FY25: $83.1k, FY26: $141k, FY27: $57.7k. Total: $281.8k ($281.8k).

2024 CSES Rapid Response R&D Grant, Planetary Science

The role of cometary impact-generated fractures on organics transport in Titan's crust.
LDRD seed funding beginning November 2024 for preliminary work and proof-of-concept results studying the effect of cometary hypervelocity impact-generated fractures on the transport of organics from Titan's surface to its subsurface ocean. Project: 20258051CT-SES. **PI: John Ortiz** (LANL); Co-Is: Esteban Rougier (LANL), Phil Stauffer (LANL). Total: $60k ($60k).

2021–2024 *CSES Student Fellowship, Planetary Science
From Manhattan to Mars: Applying models of subsurface radionuclide gas seepage from nuclear testing to understand methane release from the martian subsurface.
In-depth 3-year investigation of subsurface methane transport in fractured rock environments on Mars. Multiple gas-phase methane transport and release mechanisms were be considered in addition to more accurate prediction of atmospheric methane concentrations through subsurface-atmosphere coupling of FEHM subsurface flow & transport simulator and an atmospheric mixing model implemented in Python. Project: 20210528CR-CSE. PI: Phil Stauffer (LANL); **Student Fellow: John Ortiz** (JHU-LANL); Co-Is: Roger Wiens (LANL), Harihar Rajaram (JHU), Kevin Lewis (JHU), Anthony Toigo (JHU-APL). Total: $190k ($126k).

2020 *CSES Rapid Response R&D Grant, Planetary Science
Leveraging martian subsurface methane release data to inform models of gas release from clandestine nuclear testing.
LDRD Seed funding for preliminary work and proof-of-concept results of subsurface methane gas transport in fractured-rock environments on Mars influenced by periodic barometric pressure variations. PI: Dylan Harp (LANL); Student: **John Ortiz** (JHU-LANL); Co-Is: Roger Wiens (LANL), Harihar Rajaram (JHU), Kevin Lewis (JHU). Total: $30k ($24.7k).

Research Experience

2025 Director's Postdoctoral Fellow
National Security Earth Science Group (EES-17), Los Alamos National Laboratory
Mentor: Dr. Esteban Rougier.

> *Summary*: During my 15 months as a postdoctoral researcher in EES-17, I published two first-authored manuscripts in peer-reviewed journals, with a third accepted with revisions currently under review. I was awarded a Director's Postdoctoral Fellowship, which fully funded the final six months of my postdoc and provides an additional $200k in research funding through FY27 following my conversion to staff. I also secured an LDRD CSES Rapid Response R&D award, which gave me the opportunity to serve as a PI and mentor a PhD GRA student on a planetary science project. In addition, I convened and chaired two oral sessions at AGU. Finally, I led the integration of the ANEOS equation-of-state material model into HOSS, representing a significant new capability for high-energy physics and weapons effects research.

Director's Fellow Proposal Title: *Re-caging the Dragon: Geomechanical insights to mitigate UNE containment failure risks* (see Research Funding). Selected for Director's Postdoctoral Fellow appointment on 20 February, 2025 (see Honors & Awards), with official start date

of 7 April, 2025.

Implemented the ANEOS analytical equation of state within the HOSS material model library to add capability for material phase transitions under extreme conditions. This capability improves HOSS's ability to calculate the physics of extreme states of matter, which is critical in high-energy physics calculations, including: weapons effects, hypervelocity impacts, and cratering/ground shock.

On conversion to Director's Fellow, I used the remainder of the funding remaining from the CSES Rapid Response R&D grant (see Research Funding) to hire and serve as a co-mentor for a summer PhD student (Aditya Pujari) to carry out tasks on our Titan hypervelocity impacts project. Co-mentored PhD student with Bryan Euser and Esteban Rougier. Student presented summer work in a poster at the CSES FY25 Symposium and won the "Best Student Presentation Award".

2024 - 2025 Postdoctoral Research Associate

National Security Earth Science Group (EES-17), Los Alamos National Laboratory

Mentor: Dr. Esteban Rougier.

This postdoc appointment was from June 2024 to April 2025. Served as a PI on a successfully-funded LDRD CSES Rapid Response R&D grant proposal (see Research Funding) which pursued work modeling hypervelocity comet impacts on Titan and the potential for surface-to-ocean exchange of organic material through the generated fracture networks. This grant funded my research at 40% FTE until I was awarded the Director's Postdoctoral Fellowship.

Carried out programmatic work for government sponsors related to detecting and verifying underground nuclear explosion (UNE) tests. Performed pressure and gas transport predictions to actively support the Low Yield Nuclear Monitoring (LYNM) program, which directly fed into engineering decisions for the Physics Experiment 1 (PE1) chemical explosion field tests at the Nevada National Security Site (NNSS).

Facilitated a 4-day collaboration with members of a UC Santa Cruz hydrogeology research group (led by Distinguished Professor Dr. Andy Fisher) to assist in developing hydrothermal flow models in FEHM for investigating circulation in subsurface oceans on Europa and Enceladus as part of the NASA-funded project ExOW (Exploring Ocean Worlds; https://oceanworlds.whoi.edu/projects/exploring-ocean-worlds-exow/. Assisted in mentoring and developing modeling workflows for two current UCSC PhD students (Kristin Dickerson, Ryu Akiba). Collaboration is ongoing.

2020 - 2024 Doctoral Student, GRA (Graduate Research Assistant)

Energy and Natural Resources Security Group (EES-16), Los Alamos National Laboratory

Mentor: Dr. Philip Stauffer

Wrote two successfully-funded LDRD CSES Planetary Science grants (see Research Funding) to pursue work modeling subsurface methane transport in fractured rock at Gale crater, Mars at 70% FTE for three years. Key outcomes were two published first-authored journal articles and co-authorship of a NASA proposal with the TLS-SAM Team for multiple atmospheric sampling experiments to be performed by the MSL *Curiosity* rover, forging new ties between NASA and LANL (see Planetary Exploration Mission Experience and Research Funding). Additional outcomes include release of the Mars-specific branch of FEHM (FEHM-Mars; https://doi.org/10.5281/zenodo.10455952) that feature code modifications in Fortran to adapt to martian conditions (e.g., reduced gravity, equation-

of-state modifications consistent with Mars atmospheric "air") and a reactive transport capability allowing users to specify time-varying distribution coefficients for tracer adsorption problems related to temperature changes.

Collaborated with experimentalists in EES-14 and EES-16 to develop reactive gas transport models to determine transport properties of noble gases in variably saturated zeolitic and non-zeolitic rocks based on novel data from bench-scale diffusion experiments. Augmented reactive transport capabilities of FEHM by developing a new dual-site, competitive kinetic adsorption model to interrogate counter-intuitive radionuclide gas transport experimental results in zeolitic rocks. This work will improve gas tracer signal interpretation related to historical UNE test data and inform field-scale transport models.

Carried out programmatic work for government sponsors related to detecting and verifying underground nuclear explosion (UNE) tests. Performed pressure and gas transport predictions to support Low Yield Nuclear Monitoring (LYNM) program. Was also the lead developer Numerical Reduced Order Multiphase Model (NROMM) software package for rapid prediction of gas seepage times, which was developed for AFTAC (Air Force Technical Applications Center) as part of a Trailblazer project (see Honors & Awards). The NROMM tool is a Python wrapper for FEHM that allows the user to use a simple, readable input file to run flow & transport UNE simulations with reactive tracers while handling meshing, initialization calculations, radioactive decay, and post-processing. Led a training workshop for end users of NROMM to members of the 21st Surveillance Squadron of AFTAC (see Workshops Facilitated).

Led a NMSBA (New Mexico Small Business Assistance; https://www.nmsbaprogram.org/) project (2021-24) to help a small environmental consulting firm dramatically reduce costs associated with interpreting aquifer hydrogeologic properties from well data. Demonstrated the use of frequency domain analysis and solid-earth tidal loading to estimate hydrogeologic properties without the need for expensive pump tests.

2017 - 2019 **Post-Master's Student, GRA**
Computational Earth Science Group (EES-16), Los Alamos National Laboratory

Mentor: Dr. Dylan Harp

Lead developer of the Numerical Reduced Order Multiphase Model (NROMM) software package for rapid prediction of gas seepage times. Software application was developed for use on multiple platforms (Windows/Linux/MacOS) and was delivered to AFTAC.

Developed numerical approaches for detecting and verifying underground nuclear explosion (UNE) tests. Projects included simulating radionuclide gas transport in fractured geologic media using finite-element method (FEM) and control volume finite-element (CVFEM) numerical models, simulating high-pressure methane injection into shale samples to inform laboratory investigations, and determining laboratory- and field-scale transport properties of rocks using models and tracer experiments.

Developed a workflow for coupling discrete fracture networks (DFNs) generated in dfnWorks with a 3-D continuum mesh as a means of improving computational efficiency of flow and transport simulations related to UNEs. Developed and performed a suite of flow and transport verification tests using FEHM. Such model frameworks are useful for representing scenarios where the 3-D rock matrix or spherical cavity must be explicitly represented and coupled to planar fracture features.

Contributed to development of the Amanzi-ATS high performance computing (HPC) flow & transport simulator to meet the Nuclear Quality Assurance-1 (NQA-1) regulatory

standard by improving code verification and benchmark tests in addition to maintaining software user guide documentation. Collaborated on a multi-lab program supported by the DOE Office of Environmental Management (DOE EM) to provide scientifically defensible and standardized assessments of the uncertainties and risks associated with the environmental cleanup and closure of waste sites. This software won an R&D 100 Award in September 2020.

2016 - 2017 Graduate Research Assistant
Earth and Environmental Sciences Department, NMT

Mentor: Dr. Mark Person

Created transient 3-D finite-difference (FDM) models in MODFLOW to analyze induced seismicity risk resulting from fluid-fault interactions associated with a suite of basal reservoir injection scenarios. Also developed transient 2-D cross-sectional FDM models in MATLAB to test fluid-fault interactions for crystalline basement fault zones exhibiting local, dynamically enhanced permeability caused by excess fluid pressures. Developed new approaches for representing fault zones with multiple architectural components and identified several key hydrogeologic parameters that control deep propagation of the fluid pressure envelope and thus present increased risk of induced or triggered seismic events.

Collaborated on an NSF-funded (via NM EPSCoR) field project deploying subsurface geophysical field survey equipment (transverse electromagnetics [TEM], magnetotellurics [MT]) for interpretation of deep saline geothermal flow regimes in order to evaluate potential hydrothermal systems in southern New Mexico.

2013 REU Research Intern
College of Earth, Ocean, and Atmospheric Sciences, Oregon State University

Mentors: Dr. Peter Ruggiero, Dr. Nicholas Cohn

Completed an NSF-funded Research Experience for Undergraduates (REU) internship on coastal processes. Collected nearshore single-beam bathymetry data using jet skis. Interpolated nearshore topographic and bathymetric data to extract key spatial and temporal morphology metrics using MATLAB. Determined rates of longshore-uniform sandbar migration cycles along the Oregon and Washington coasts representing a huge component of seasonal coastal sediment flux.

2013 Research Intern
United States Army Corps of Engineers Field Research Facility, Duck NC

Mentors: Dr. Kate Brodie, Dr. Heidi Wadman

Supported active coastal evolution and erosion research by collecting and processing LiDAR observations during and after storm surges using both beach-based and amphibious vehicles (LARC-V; Lighter Amphibious Resupply Cargo 5-ton).

Created a paleo-hurricane record of St. Croix by performing grain-size analysis on sediment cores combined with charcoal carbon dating.

Workshops & Shortcourses

Workshops Facilitated

2025 FEHM Workshop

Presented at FEHM workshop led by Dr. Philip Stauffer. Demonstrated capabilities relevant to multi-phase reactive contaminant transport, high-pressure/high-temperature flow simulations, and barometric pumping.

2021 NROMM Demo and Training for AFTAC

Led a training workshop and software demonstration for members of the AFTAC 21st Surveillance Squadron on use of the NROMM software package (see Honors & Awards). The NROMM tool I developed is a Python wrapper for FEHM that allows the user to use a simple, readable input file to run flow & transport UNE simulations with reactive tracers while handling meshing, initialization calculations, radioactive decay, and post-processing.

Relevant Shortcourses and Workshops Attended

2026 LYNM Gas Transport Modeling Workshop

Pacific Northwest National Laboratory, Richland, WA

Attended 2-day quad-lab workshop at PNNL for lessons-learned discussions related to completed work and forward-planning of predictions efforts for the LYNM Physics Experiment 1 (PE1) venture. Presented on behalf of LANL (see Conference & Meetings Talks) detailing modeling work completed for PE1-A and predictions for future experiments in the series.

2023 LYNM Underground Nuclear Weapons Testing Orientation Program (UNWTOP)

Nevada National Security Site (NNSS), Las Vegas, NV

Attended 4-day multi-lab program at the NNSS for nuclear intelligence orientation regarding history, challenges, and scientific topics related to underground nuclear weapons testing.

Teaching Experience

2019 Grading Assistant

Environmental Health & Engineering Department, Johns Hopkins University

Graded weekly homework assignments for 20+ upper-level undergraduate engineering students in an Introduction to Fluid Mechanics course. Occasionally taught course lectures as needed.

2015 - 2016 Graduate Teaching Assistant

Earth and Environmental Sciences Department, NMT

Developed lesson plans and led class lectures and field trips for 20+ students in intermediate and upper level undergraduate Earth Sciences lab sections. Taught lab sections for courses in both Geomorphology and Stratigraphy & Paleontology with work load totaling 20 hrs/week.

Skills

Programming/command languages

Python	Fortran	Matlab	R
C++	LaTeX	Bash/shell	

Software, codes

FEHM	PFLOTRAN	HOSS	LaGriT
Cubit	DFNWorks	Git	ParaView
GDB	MODFLOW	OpenFOAM	SPOTL
Sphinx	Adobe Illustrator		

Publications & Presentations

Total Citations: 290
h-index: 11
1st-author peer-reviewed journal articles: 6

(Citation data from Google Scholar, current as of 8 September, 2026).

Preprints

2026 Fernando, B., Heldmann, J., Grey, B., **Ortiz, J. P.**, Euser, B., Seligman, D., Kim, E., Colaprete, A., Alessi, E. M., Koschny, D., Cook, A., Green, J., King, P., Teng, S., Graninger, D., Goldberg, A., Cooke, W., Skrutskie, M. F., Schlaufman, K., Schmerr, N., Donahue, C. M., Schmidt, C. A., Chanover, N. J. Observational planning for the August 5, 2026 Falcon 9 Upper Stage lunar impact. (2026). *arXiv*. doi: 10.48550/arXiv.2607.14625.

Under Review, Submitted, & In Prep

Under Review **Ortiz, J. P.**, Lucero, D. D., Fritz, B. G., Neil, C. W., Boukhalfa, H., Stauffer, P. H., The PE1 Team. Long-term gas migration for an underground chemical explosive test: A comparison of numerical simulations to multiyear borehole data. (2026). *Scientific Reports*. (Under Review).

Under Review Lucero, D. D., Sweeney, M. R., Hyman, J. D., Zyvoloski, G. A., Miller, T. A., Hinrichs, E. L., Dalal, M. V., **Ortiz, J. P.**, Stauffer, P. H. Modeling Inertial Laminar Flow in Fractured Media: Network-Scale Consequences (2026). *Transport in Porous Media*. (Under Review).

Submitted Fernando, B., Heldmann, J., Grey, B., **Ortiz, J. P.**, Euser, B., Seligman, D., Kim, E., Colaprete, A., Alessi, E. M., Koschny, D., Cook, A., Green, J., King, P., Teng, S., Graninger, D., Goldberg, A., Cooke, W., Skrutskie, M. F., Schlaufman, K., Schmerr, N., Donahue, C. M., Schmidt, C. A., Chanover, N. J. Observational planning for the August 5, 2026 Falcon 9 Upper Stage lunar impact. (2026). *The Astrophysical Journal Letters*. (Submitted). doi: 10.48550/arXiv.2607.14625.

In Prep. **Ortiz, J. P.**, Pujari, A. R., Euser, B., Rougier, E. Modeling fracture and melt generation from a hypervelocity comet impact on Titan. (2026). *International Journal of Impact Engineering*. (In Prep).

In Prep. Morris, J., Foxe, M. P., Churby, A., Knox, H., Myers, S., Stauffer, P. H., Lucero, D. D., **Ortiz, J. P.**, et al. Overview of the Interdisciplinary Physics Experiment 1 in Support of Low Yield Nuclear Monitoring. (2026). *Rock Mechanics and Rock Engineering*. (In Prep).

Journal articles & Reports

Peer reviewed

2026 Neil, C. W., Eldridge, D. L., Miller, H. D., **Ortiz, J. P.**, Fritz, B. G., Euler, G. G., Stauffer, P. H., The PE1 Team. Linking Laboratory and Field Observations of Carbon Dioxide Transport at the Nevada National Security Site. (2026). *ACS Omega*. doi:10.1021/acsomega.6c02107. (Selected as Editor's Choice; see Honors & Awards).

2026 **Ortiz, J. P.**, Lucero, D. D., Rougier, E., Knight, E. E., Bourret, S. M., Fritz, B. G., Bodmer, M. A., Heath, J., Neil, C. W., Boukhalfa, H., Kuhlman, K., Otto, S., Ezzedine, S., Roberts, B. L., Choens, R. C., Zyvoloski, G.A., Stauffer, P. H., The PE1 Team. Predicting multiphase flow and tracer transport for an underground chemical explosive test. (2026). *Scientific Reports*. 16. doi:10.1038/s41598-026-35868-w. (Cited by: 1).

2025 **Ortiz, J. P.**, Lewis, K. W., Wiens, R. C., Stauffer, P. S., Lewis, K. W, Harp, D. R., Rajaram, H. Modeling barometric pumping of martian methane: Implications for *Perseverance* sample timing and gas loss from drilled cores. (2025). *Icarus*. 441, 116702. doi:10.1016/j.icarus.2025.116702.

2025 **Ortiz, J. P.**, Neil, C. W., Rajaram, H., Boukhalfa, H., Stauffer, P. S. Preferential adsorption of noble gases in zeolitic tuff with variable saturation: A modeling study of counter-intuitive diffusive-adsorptive behavior. (2025). *Journal of Environmental Radioactivity*. 282, 107608. doi:10.1016/j.jenvrad.2024.107608. (Cited by: 9).

2025 Neil, C. W., Eldridge, D. L., Miller, H., **Ortiz, J. P.**, Stauffer, P. H., Rahn, T., Broome, S. T., Boukhalfa, H., Euler, G. G. Explosive byproduct gas transport through sorptive geomedia. (2025). *Transport in Porous Media*. 152, 80. doi:10.1007/s11242-025-02212-1. (Cited by: 4).

2025 Lucero, D. D., Bourret, S. M., **Ortiz, J. P.**, Fritz, B. G., Bodmer, M. A., Heath, J., Kuhlman, K., Otto, S., Ezzedine, S., Robert, B. L., Choens, R. C., Person, M. A., Stauffer, P. S., The PE1 Team. (2025). Permeability scaling relationships from core- to field-scale measurements: A characterization of P-Tunnel, Nevada National Security Site. *Scientific Reports*. 15(1), 12938. doi:10.1038/s41598-025-96835-5. (Cited by: 7).

2024 Neil, C. W., Swager, K. C., Bourret, S. M., **Ortiz, J. P.**, Stauffer, P. S. Rethinking porosity-based diffusivity estimates for sorptive gas transport at variable temperatures. (2024). *Environmental Science & Technology*. doi:10.1021/acs.est.4c04048 (Cited by: 11).

2024 **Ortiz, J. P.**, Rajaram, H., Stauffer, P. S., Lewis, K. W, Wiens, R. C., Harp, D. R. Subdiurnal methane variations on Mars driven by barometric pumping and planetary boundary layer evolution. (2024). *Journal of Geophysical Research: Planets*. 129, e2023JE008043. doi:10.1029/2023JE008043. (Cited by: 7).

2022 **Ortiz, J. P.**, Rajaram, H., Stauffer, P. S., Harp, D. R., Wiens, R. C., Lewis, K. W. Barometric pumping through fractured rock: a mechanism for venting deep methane to Mars' atmosphere. (2022). *Geophysical Research Letters*. doi:10.1029/2022GL098946. (Cited by: 15).
(Cover article: https://agupubs.onlinelibrary.wiley.com/doi/epdf/10.1002/grl.62460).

2022 Neil, C. W., Boukhalfa, H., Xu, H., Ware, S. D., **Ortiz, J. P.**, Avendaño, S. T., Harp, D. R., Broome, S., Hjelm, R. P., Roback, R., Brug, W. P., Stauffer, P. H. Gas diffusion through variably-water-saturated zeolitic tuff: implications for transport following a subsurface nuclear event. (2022). *Journal of Environmental Radioactivity*. doi:10.1016/j.jenvrad.2022.106905. (Cited by: 22).

2021 Avendaño, S. T., Harp, D. R., Kurwadkar, S., **Ortiz, J. P.**, Stauffer, P. H. Continental-scale geographic trends in barometric-pumping efficiency potential: a North American case

study. (2021). *Geophysical Research Letters*. doi:10.1029/2021GL093875. (Cited by: 12).

2020 Petrie, E. S., Bradbury, K. K., Cuccio, L., Smith, K., Evans, J. P., **Ortiz, J. P.**, Kerner, K., Person, M. A., Mozley, P. S. Geologic characterization of nonconformities using outcrop and whole-rock core analogues: hydrologic implications for injection-induced seismicity. (2020). *Solid Earth*, 11(5),1803-1821. doi:10.5194/se-11-1803-2020.

2020 Bourret, S. M., Kwicklis, E. M., Harp, D. R., **Ortiz, J. P.**, Stauffer, P. H. Beyond Barnwell: Applying lessons learned from the Barnwell site to other historic underground nuclear tests at Pahute Mesa to understand radioactive gas-seepage observations. (2020). *Journal of Environmental Radioactivity*, 222. doi:10.1016/j.jenvrad.2020.106297. (Cited by: 28).

2019 Harp, D. R., **Ortiz, J. P.**, Stauffer, P. H. Identification of dominant gas transport frequencies during barometric pumping of fractured rock. (2019). *Scientific Reports (Nature Publishing Group)*, 9(1), 9537. doi:10.1038/s41598-019-46023-z. (Cited by: 32).

2019 **Ortiz, J. P.**, Person, M. A., Mozley, P. S., Evans, J. P., Bilek, S. L. The role of fault-zone architectural elements on pore pressure propagation and induced seismicity. (2019). *Groundwater*, 57(3): 465-478. doi:10.1111/gwat.12818. (Cited by: 28).

2019 Stauffer, P. H., Rahn, T., **Ortiz, J. P.**, Salazar, L. J., Boukhalfa, H., Behar, H. R., Snyder, E. E. Evidence for High Rates of Gas Transport in the Deep Subsurface. (2019). *Geophysical Research Letters*. doi:10.1029/2019GL082394. (Cited by: 25).

2018 Harp, D. R., **Ortiz, J. P.**, Pandey, S., Karra, S., Anderson, D., Bradley, C., Viswanathan, H., & Stauffer, P. H. Immobile Pore-Water Storage Enhancement and Retardation of Gas Transport in Fractured Rock. (2018). *Transport in Porous Media*, 1-26. doi:10.1007/s11242-018-1072-8. (Cited by: 32).

2017 **Ortiz, J. P.** The Role of Fault-Zone Architectural Elements and Basal Altered Zones on Downward Pore Pressure Propagation and Induced Seismicity in the Crystalline Basement. (2017). [Master's Thesis] *New Mexico Institute of Mining and Technology*. (Cited by: 1).

2016 Zhang, Y., Edel, S. S., Pepin, J., Person, M., Broadhead, R., **Ortiz, J. P.**, Bilek, S. L., Mozley, P. S., & Evans, J. P. Exploring the potential linkages between oil-field brine reinjection, crystalline basement permeability, and triggered seismicity for the Dagger Draw Oil field, southeastern New Mexico, USA, using hydrologic modeling. (2016). *Geofluids*. doi:10.1111/gfl.12199 (Cited by: 20).

2014 **Ortiz, J. P.** Quantifying regional sediment flux from observations of nearshore morphology in the Columbia River Littoral Cell. (2014). [Undergraduate Thesis] *Dartmouth College Senior Honors Thesis Collection*.

2014 Cohn, N., Ruggiero, P., **Ortiz, J. P.**, & Walstra, D. J. Investigating the role of complex sandbar morphology on nearshore hydrodynamics. (2014). *Journal of Coastal Research*, 70(sp1), 53-58. doi:10.2112/SI65-010.1. (Cited by: 27).

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2018 Stauffer, P. H., Rahn, T. A., **Ortiz, J. P.**, Salazar, L. J., Boukhalfa, H., & Snyder, E. E. Summary of a Gas Transport Tracer Test in the Deep Cerros Del Rio Basalts, Mesita del Buey, Los Alamos NM. United States. (2018). doi:10.2172/1417180. (Cited by: 2).

Conference & Meetings Talks

2026 "From Manhattan to Mars: Applying Models of Subsurface Radionuclide Gas Seepage From Nuclear Testing to Understand a Martian Methane Mystery", 75th Lindau Nobel Laureate Meeting; *Next Gen Science Presentations*. Lindau, Germany. 1 July, 2026.

(see Lindau Meeting Programme)

2026 "PE1-B and -DL cavity & near field predictions", Physics Experiment 1: B and DL Predictions Meeting. Pacific Northwest National Laboratory, Richland, WA. 30 July, 2026.

2026 "LANL – GS Borehole pressure and concentration predictions", Physics Experiment 1: B and DL Predictions Meeting. Pacific Northwest National Laboratory, Richland, WA. 28 July, 2026.

2026 "Long-term Gas Migration Following PE1-A: A Comparison of Numerical Simulations to Multiyear Borehole Data", LYNM Quad-Laboratory All-Hands Meeting. Lawrence Livermore National Laboratory, Livermore, CA. 18 June, 2026.

2026 "Sequential Kinetic-Explosive Penetrators for Rapid-Response Lunar and Cislunar Defense", Lunar Townhall. Los Alamos National Laboratory. 26 February, 2026.

2026 "LANL PE1 Flow & Transport Predictions", LYNM Gas Transport Modeling Workshop. Pacific Northwest National Laboratory, Richland, WA. 21 January, 2026. (see Relevant Shortcourses & Workshops Attended).

2025 "Cometary impact-generated fractures as potential pathways for surface-to-ocean exchange of organics on Titan". *Session: P32A Ice and Ocean Worlds: Geology, Oceanography, Chemistry, and Habitability IV Oral*. AGU 2025 Fall Meeting in New Orleans, CA. 17 December, 2025.

2024 "Preferential adsorption of xenon in variably-saturated zeolitic tuff", LYNM Quad-Laboratory All-Hands Meeting. Lawrence Livermore National Laboratory, Livermore, CA. 25 June, 2024.

2023 "Sub-diurnal methane variations on Mars driven by barometric pumping and planetary boundary layer evolution". *Session: P44C The New Mars Underground III Oral*. AGU 2023 Fall Meeting in San Francisco, CA. 14 December, 2023.

2023 "The Mars Underground: Characterizing methane seepage on the Red Planet", LANL Center for Space and Earth Science (CSES) Symposium. 23 August, 2023.

2023 "The Mars Underground: Characterizing methane seepage on the Red Planet", LDRD Appraisal for FY21-23, CSES Planetary Science Student Fellow. 17 April, 2023.

2023 "NROMM: Numerical Reduced-Order Multiphase Model | A tool for making rapid predictions of UNE gas seepage", LYNM Program Technical Meeting. Los Alamos National Laboratory, Los Alamos, NM. 23 January, 2023.

2022 "Using legacy radionuclide data to validate barometric pumping models", LYNM Quad-Laboratory All-Hands Meeting. Sandia National Laboratory, Albuquerque, NM. 25 May, 2022.

2022 "PE1-A pressure and permeability predictions", LYNM Quad-Laboratory All-Hands Meeting, Sandia National Laboratory. 26 May, 2022.

2021 "PE1-A pressure and gas arrival predictions based on in-situ permeability measurements", Nuclear Test Monitoring Exchange of Information by Visit and Report (EIVR) 58. 12 October, 2021.

2018 "Analysis of enhanced gas transport in fractured rock due to barometric pressure variations" (presenting for Dylan Harp). InterPore 10th Annual Meeting in New Orleans, LA. May 2018.

2017 "Improving estimates of subsurface gas transport in unsaturated fractured media using field tracer data and numerical methods." *Session: H52A Advances in Hydrological Characterization of Flow and Transport in Fractured Media: Numerical and Experimental Observations II*. AGU 2017 Fall Meeting in New Orleans, LA. 15 December 2017.

2016 "The Hydrologic Connection Between Basal Reservoir Injection, Crystalline Basement Fault

Zones, and Induced Seismicity". GSA Annual Meeting in Denver, CO. September 2016. (Cited by: 1).

Conference Posters

2026 Leveraging Geoscience Simulation Tools for Insights into Planetary Exploration. *LANL Engineers Week*. Los Alamos National Laboratory. Los Alamos, NM. 25 February, 2026.

2025 Predicting multiphase flow and tracer transport for an underground chemical explosive test. *Session: H13Z: Upscaling of Multiphysical Transport in Porous Media: Advancements in Analytical, Numerical, Data-Driven, and Symbolic Computational Methods for Scale Translation Poster*. AGU 2025 Fall Meeting in New Orleans, LA. 15 December, 2025.

2025 Modeling barometric pumping of martian methane: Implications for Perseverance sample timing and gas loss from drilled cores. *Session: P21H The New Mars Underground VIII: Maximizing Martian Science Return Poster*. AGU 2025 Fall Meeting in New Orleans, LA. 16 December, 2025.

2024 The effect of variable saturation on preferential adsorption of xenon in zeolitic tuff: A two-site kinetic competitive adsorption model. *Session: H53L - Reactive Transport and Chemo-mechanical Processes in Porous Media II Poster*. AGU 2024 Fall Meeting in Washington D.C. (Cited by: 1).

2024 Planetary Boundary Layer constraints on sub-diurnal methane variations on Mars driven by barometric pumping: A case for strategic timing of Perseverance regolith samples. *Session: P13E The New Mars Underground: Fluids, Volatiles, and the Future of Mars Exploration II Poster*. AGU 2024 Fall Meeting in Washington D.C. (Cited by: 1).

2024 From Manhattan to Mars: Applying Models of Subsurface Radionuclide Gas Seepage from Nuclear Testing to Understand Methane Release from the Martian Subsurface. *LANL Center for Space and Earth Science (CSES) Symposium*. 7 August, 2024.

2023 Sub-diurnal methane variations on Mars driven by barometric pumping and planetary boundary layer evolution. *LANL Student Symposium*.

2022 Barometric pumping through fractured rock: A mechanism for venting deep underground methane to Mars' atmosphere. *Session: P22F The New Mars Underground: Nexus of Decadal Planetary Science Objectives II Poster*. AGU 2022 Fall Meeting in Chicago, IL.

2018 A reduced-order model to assist real-time predictions of gas transport in unsaturated fractured media. InterPore 10th Annual Meeting in New Orleans, LA. May 2018.

2018 Coupled discrete fracture and 3D continuum domain representation to efficiently capture gas transport from underground cavities. *Session: H51P Coupled Processes in Fractured Media Across Scales: Experimental and Modeling Advances Posters*. AGU 2018 Fall Meeting in Washington D.C.

2018 Identifying dominant barometric frequencies driving gas transport in fractured rock (presenting for Dylan Harp). *Session: H51P Coupled Processes in Fractured Media Across Scales: Experimental and Modeling Advances Posters*. AGU 2018 Fall Meeting in Washington D.C.

2013 Interannual Sandbar Variability within the Columbia River Littoral Cell. *Session: EP13A Coastal Geomorphology and Morphodynamics I Posters*. AGU 2013 Fall Meeting in San Francisco, CA.

News Coverage/Web Articles

2026 "4 Los Alamos scientists to meet with Nobel laureates". *STE Highlights*, 24 April 2026. https://www.lanl.gov/media/newsletters/ste-highlights/04-2026-lindau-meeting.

2026 "A SpaceX Rocket is Headed for a Crash on the Moon". *The New York Times*, 4 August 2026. https://www.nytimes.com/2026/08/04/science/spacex-rocket-moon-crash.html.

2026 "'Spectral lines' and a giant plume: what science saw when the SpaceX rocket hit the moon". *The Guardian*, 6 August 2026. https://www.theguardian.com/science/2026/aug/06/spacex-rocket-crash-into-moon-impact.

2026 "Monitoring a lunar impact in real time". *LANL Home*, 5 August 2026. https://www.lanl.gov/media/news/0805-lunar-impact.

2024 "Mystery of Mars' 'Burps' Could Aid Search for Life". ***Newsweek***, 25 January 2024. https://www.newsweek.com/mystery-mars-burp-belch-methane-search-life-1863907.

2025 "Congratulations to our FY25 CSES Symposium Student Presentation Award winners!". *CSES Newsletter*, 9 September 2025. https://int.lanl.gov/news/newsletters/cses/september-2025/index.html.

2024 "Science Highlights: Simulations help Curiosity search for signs of past or present life on Mars". *CSES Newsletter*, 2 April 2024. https://int.lanl.gov/news/newsletters/cses/april-2024/index.html.

2024 "Study Predicts Best Times for Rover to Sample Mars Methane in Search for Life". *Johns Hopkins University Engineering News*, 26 January 2024. https://engineering.jhu.edu/news/study-predicts-best-times-for-rover-to-sample-mars-methane-in-search-for-life/.

2024 "Atmospheric Pressure Changes Could Be Driving Mars' Elusive Methane Pulses". *Los Alamos Daily Post*, 26 January 2024. https://ladailypost.com/lanl-atmospheric-pressure-changes-could-be-driving-mars-elusive-methane-pulses/.

2024 "Methane pulses on Mars possibly driven by atmospheric pressure changes". *Phys.org*, 24 January 2024. https://phys.org/news/2024-01-methane-pulses-mars-possibly-driven.html.

2024 "Mars' methane release tied to atmospheric pressure fluctuations". *New Mexico Sun*, 25 January 2024. https://newmexicosun.com/stories/653884854-mars-methane-release-tied-to-atmospheric-pressure-fluctuations.

2024 "Mars' methane release tied to atmospheric pressure fluctuations". *Türkiye Newspaper*, 25 January 2024. https://www.turkiyenewspaper.com/science/17875.

2024 "火星でメタンを測るには日の出の直前が最適？メタン濃度の変化をモデル計算で予測 [Is the best time to measure methane on Mars just before sunrise? Predicting changes in methane concentration using model calculations]". *Sorae*, 12 February 2024. https://sorae.info/astronomy/20240212-mars-methane.html

2024 "Atmospheric pressure changes could be driving Mars' elusive methane pulses". *LANL News Stories*, 24 January 2024. https://discover.lanl.gov/news/0124-mars-methane-pulses/.

2024 "Atmospheric pressure changes could be driving Mars' elusive methane pulses". *LANL News Stories*, 24 January 2024. https://int.lanl.gov/news/news_stories/2024/january/0124-mars-methane-pulses.shtml?source=topnews (internal LANL webpage).

2024 "Atmospheric pressure changes could explain Mars methane." *Universe Today*, 29 January 2024. https://www.universetoday.com/165470/atmosphere-pressure-changes-could-explain-mars-methane/.

2024 "NASA's Curiosity Rover Closer to Solving Mystery of Methane Biosignature on Mars, Aiding Search for Life". *The Debrief*, 5 February 2024. https://thedebrief.org/nasas-curiosity-rover-closer-to-solving-mystery-of-methane-biosignature-on-mars-aiding-search-for-life/.

2022 "Study illuminates Mars methane transmission from subsurface depths that could indicate microbial source". *STE Highlights*, August 2022. https://www.lanl.gov/science-innovation/science-highlights/2022/2022-08.php#EarthandEnvironmentalSciences-1

2022 "Study illuminates Mars methane transmission from subsurface". *LANL News Stories*, 21 August 2022. https://int.lanl.gov/news/news_stories/2022/august/0822-mars-methane.shtml

(internal LANL webpage).

2022 "Progress on understanding radioactive gas migration from underground nuclear explosions". *STE Highlights*, February 2022. https://www.lanl.gov/science-innovation/science-highlights/2022/2022-02.php#EarthandEnvironmentalSciences-1

Departmental Seminars and Other Talks

2023 "Predicting martian methane variations to identify strategic sampling times for the *Curiosity* rover", LANL, EES-16 SFT Lightning Talk. 1 November, 2023.

2022 "The Mars Underground: Characterizing subsurface methane seepage on the Red Planet", Johns Hopkins University, Baltimore, MD. Environmental Health & Engineering Seminar. 20 September 2022.

2021 "From Manhattan to Mars: Applying models of subsurface radionuclide gas seepage from nuclear testing to understand methane release from the martian subsurface", Johns Hopkins University, Baltimore, MD. Environmental Health & Engineering Seminar. 2 November 2021.

2021 "Determining hydrogeologic properties using well data, barometric pressures, and tidal analysis", LANL, EES-16 Science Café series. 12 August 2021.

2021 "From Manhattan to Mars: Generating novel insights into methane fluctuations on the Red Planet", Johns Hopkins University, Baltimore, MD. Environmental Health & Engineering Seminar. 2 March 2021.

2019 "Improving estimates of gas transport in fractured rock – Implications for verification of underground nuclear events", Johns Hopkins University, Baltimore, MD. Environmental Health & Engineering Seminar. 29 October 2019.

2020 "NROMM Seepage Tool: recent capability development and analytical verification", LANL, EES-16 Science Café series. 23 July 2020.

2019 "Noble gas diffusion through variably saturated rock – implications for verification of subsurface nuclear events", LANL, EES-16 Science Café series. 8 August 2019.

2018 "The role of fault-zone architectural elements and basement altered zones on pore pressure propagation and induced seismicity", LANL, EES-16 Science Café series. 2 August 2018.

2017 "Improving estimates of subsurface gas transport in unsaturated fractured media using field tracer data and numerical methods", LANL, EES-16 Science Café series. 30 November 2017.

Professional Affiliations

2013 - pres. American Geophysical Union (AGU)

2015 - pres. American Association of Petroleum Geologists (AAPG)

2016 - pres. Geological Society of America (GSA)

2017 - pres. National Ground Water Association (NGWA)

2017 - pres. New Mexico Geological Society (NMGS)

2018 - pres. International Society for Porous Media (InterPore)